

# Ranking Transfers, Thresholds Don’t: Unsupervised Test-Time Adaptation for Cross-Corpus Audio Deepfake Detection

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**Abstract**—Audio deepfake detectors generalise poorly across corpora despite near-perfect in-domain accuracy. We show that a fine-tuned self-supervised detector’s *ranking* (ROC-AUC) largely transfers cross-corpus even when its decision *threshold* does not, and build an unsupervised *test-time adaptation* (TTA) method on that asymmetry: self-train on confident-ranked pseudo-labels from the extreme quantiles of unlabeled target scores, enforce channel-consistency, and adapt only LayerNorm and the classifier head—no target labels, no source data. On a four-target, ten-seed leave-one-corpus-out protocol, EER improves on three targets (In-the-Wild 12.78  $\rightarrow$  11.33%, Arabic 22.50  $\rightarrow$  20.99%, ASVspoof2019 5.03  $\rightarrow$  3.47%; two survive Holm correction) and is neutral on the fourth, the one with the weakest source ranking. Applied unmodified to nine *publicly released* checkpoints we did not train—five architectures, one a spectrogram transformer—over 369 cross-corpus cells, it closes the gap between accuracy at the shipped threshold and the accuracy the ranking already supports (7.3  $\rightarrow$  2.0 points,  $p \approx 10^{-17}$ , adapting 16,706 parameters) while its effect on ranking is an order of magnitude weaker (+0.9 EER, and split by corpus): the mechanism is threshold repair, and a label-free median-threshold rule matches the full method on average—though where source ranking is strong the gradient step does add per-attack separability a threshold cannot, and label-free prior estimation lifts most of the class-balance requirement. The EER gains that do occur track *calibration headroom*: on two corpora sharing no lineage with the source pool (WaveFake, 2024 commercial TTS) adaptation improves EER by +2.5 and +2.9 points ( $p=0.002$ ), larger than on the original targets. Entropy-minimisation TTA (Tent) is instead unpredictably catastrophic, and a non-SSL backbone never reaches a ranking worth adapting.

**Index Terms**—audio deepfake detection, anti-spoofing, test-time adaptation, cross-corpus generalisation, self-supervised learning, EER.

## I. INTRODUCTION

Self-supervised (SSL) front-ends such as wav2vec 2.0 and XLS-R, paired with lightweight back-ends, reach very low in-domain error on standard anti-spoofing benchmarks [1]–[3]. Deployed on audio from a different source—new microphones, codecs, languages, or generators—accuracy drops steeply, a well-documented generalisation gap [5], [6], [12].

We start from a diagnostic measurement: a fine-tuned XLS-R detector evaluated on an unseen corpus keeps a high ROC-AUC while its accuracy at a fixed threshold collapses—it still *ranks* clips correctly but its operating point is miscalibrated. Because ranking transfers, its most-confident pre-

dictions on unlabeled target audio are trustworthy enough to adapt on, *at test time*, without any labels.

Our contributions: (i) an operational use of the ranking-calibration asymmetry—ranking transfers cross-corpus even when the threshold does not—as the precondition that makes unsupervised self-training viable; (ii) a simple, stable TTA method—confident-ranked pseudo-label self-training plus channel-consistency, adapting only LayerNorm and the head; (iii) a four-target, multilingual, ten-seed leave-one-corpus-out evaluation against seven comparison points (source-only, Tent, BatchNorm-only recalibration, self-training-only, a non-SSL backbone, DANN, and an ASDG-style aggregation-separation baseline), including the one target where the method does not help; (iv) a mechanistic analysis—proxy  $\mathcal{A}$ -distance, a label-free median-threshold control that matches the full method on average, a per-attack test showing the gradient step *does* add separability where source ranking is strong, and a class-balance study with a label-free prevalence estimator that removes most of the balance requirement—placing the mechanism at *operating-point repair*; and (v) evidence that the method acts as a *label-free wrapper for third-party detectors*, restoring the operating point of nine publicly released checkpoints across five architectures and ten corpora (369 cells) without retraining them.

**Related work.** Frozen or fine-tuned SSL front-ends with AASIST or attentive back-ends dominate recent benchmarks [1]–[4]; we additionally test whether the cross-corpus gap is an SSL question at all via a from-scratch, non-SSL backbone. Train-time domain generalisation (DG) methods align real and separate fake across source domains (aggregation-separation [5]) or run gradient-reversal domain-adversarial training over source and target jointly (DANN) [8]; both act at training time rather than at deployment. Entropy minimisation (Tent) [7] is the standard TTA recipe in vision but, as we show, is unstable here; batch-statistics recalibration [9] is a no-op for LayerNorm architectures. We borrow the proxy  $\mathcal{A}$ -distance from Ben-David et al. [10] to test mechanism.

## II. METHOD

Let  $x$  be a waveform with label  $y \in \{0, 1\}$  (fake=1). A detector  $f_\theta$  pools frame features into an embedding  $e$  and

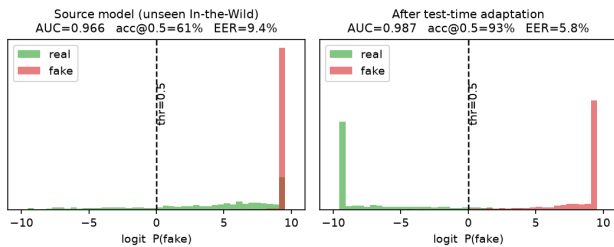


Fig. 1. Detector scores on unseen In-the-Wild audio. **Left:** the source model pushes many real clips past  $\tau=0.5$  (accuracy well below AUC) yet still ranks most real below fake. **Right:** TTA shifts the real mass back across the threshold.

outputs  $s_\theta(x) = \sigma(w^\top e + b) = P_\theta(y=1 \mid x)$ : XLS-R-300M with only its top four transformer layers trainable, attentive temporal pooling, and a linear head. Its normalisation is *LayerNorm*, which carries no running batch statistics, so a batch-statistics-recalibration baseline is architecturally a no-op here and confidence-driven gradient adaptation is the minimal mechanism that can move these layers at all.

**The ranking-calibration gap.** Ranking quality  $AUC = \Pr[s_\theta(x^+) > s_\theta(x^-)]$  is threshold-free and invariant to reparametrisation of  $s_\theta$ ; accuracy at a fixed threshold  $\tau$  is not. Empirically, AUC degrades far less than accuracy at  $\tau=0.5$  under source  $\rightarrow$  target shift (Fig. 1). We therefore trust score *order*: the extreme quantiles of unlabeled target scores are higher-precision pseudo-labels than a fixed threshold, even when  $s_\theta$  is mis-calibrated. The fixed-quantile split  $Q_q$  vs.  $Q_{1-q}$  implicitly assumes both a reliable source ranking and roughly balanced target classes; we scope the method to that regime, and Sec. IV and Sec. V show what happens when each assumption breaks.

**Test-time adaptation.** Each epoch, we assign confident pseudo-labels from the  $q$ - and  $(1-q)$ -quantiles of target scores, ignoring the middle; self-train with cross-entropy on this confident set ( $\mathcal{L}_{ST}$ ); and enforce prediction invariance under a stochastic channel perturbation  $a(\cdot)$  (gain+noise) via an MSE consistency term  $\mathcal{L}_{cons}$  between clean and perturbed views (stop-gradient on the clean target). The objective is

$$\min_{\{\gamma, \beta\}, \phi} \mathcal{L}_{ST} + \lambda \mathcal{L}_{cons}, \quad \lambda=0.3, q=0.3, E=4, \quad (1)$$

optimising *only* the LayerNorm affine parameters  $\{\gamma, \beta\}$  of the fine-tuned layers and the head  $\phi$ ; the rest stays frozen. Restricting the trainable set and using only confident tails prevents the degenerate collapse to which entropy minimisation is prone.

**Domain divergence.** To test whether adaptation works by making source and target indistinguishable in embedding space (as adversarial DG methods explicitly optimise for), we compute a proxy  $\mathcal{A}$ -distance  $\hat{d}_A = 2(1-2\epsilon_h)$  following Ben-David et al. [10], where  $\epsilon_h$  is a linear domain classifier’s held-out error on frozen embeddings, before and after adaptation.

### III. EXPERIMENTAL SETUP

**Protocol.** Leave-one-corpus-out over four EER-capable targets: ASVspoof2019-LA [13]; D2 (LibriSpeech bona fide

TABLE I  
CROSS-CORPUS EER % / AUC, MEAN OVER 10 SEEDS  
(LEAVE-ONE-CORPUS-OUT, TRANSDUCTIVE). BEST OF {SOURCE, TENT, ST-ONLY, OURS} PER COLUMN IN **BOLD**.

| Method                     | ASVspoof2019             | LibriSpeech-TTS         | In-the-Wild              | Arabic                  |
|----------------------------|--------------------------|-------------------------|--------------------------|-------------------------|
| Source-only                | 5.03 / .989              | 33.54 / .727            | 12.78 / .945             | 22.50 / .843            |
| Tent [7]                   | 23.47 $\pm$ 21.96 / .776 | 35.44 $\pm$ 3.20 / .674 | 27.34 $\pm$ 11.21 / .735 | 47.35 $\pm$ 5.03 / .527 |
| Self-training only         | 4.20 / .987              | <b>33.39</b> / .709     | 11.55 / .950             | 22.19 / .846            |
| <b>Ours (TTA)</b>          | <b>3.47</b> / .994       | 33.74 / .727            | <b>11.33</b> / .958      | <b>20.99</b> / .866     |
| Oracle (target-supervised) | 0.89 / .999              | 13.93 / .939            | 4.42 / .992              | 9.50 / .968             |

speech vs. fakes from eight open TTS/voice-cloning systems); In-the-Wild [12]; and the Arabic Audio Deepfake set (ArAD) [15]. For each held-out target the source pool merges the other three corpora with a 38-language MLAAD subset [14] (fake-only); held-out target audio, including its MLAAD-overlapping languages, is never in any source pool, and six MLAAD languages are held out of every pool to test unseen-language generalisation.

**Comparison points.** Source-only; Tent [7] (entropy minimisation, same trainable set); BN-only (AdaBN [9], no gradient step); self-training-only ( $\lambda=0$ ); Oracle (target-supervised upper bound); RawNet2Lite (from-scratch non-SSL backbone [4], trained identically); DANN [8], trained to convergence with the held-out target’s *unlabeled* inputs through a gradient-reversal branch—more privileged access than ours; and an ASDG-style aggregation-separation baseline [5], which aligns across *source* domains only.

**Implementation.** 16 kHz audio, 3 s crops, top-4-layer XLS-R fine-tuning, Adam  $1r2 \times 10^{-4}$ , batch 32, source trained 8 epochs. Adaptation: same batch and crop,  $1r1 \times 10^{-4}$ ,  $E=4$  passes;  $a(\cdot)$  is per-clip gain  $g \sim \mathcal{U}(0.7, 1.3)$  plus Gaussian noise ( $\sigma=0.005$ ); the head carries over from the source model. Tables II and I and every significance test use ten seeds per target for every method; the third-party,  $\mathcal{A}$ -distance and 2021-DF studies give their run counts inline. Code and manifests: [URL withheld].

### IV. RESULTS

Table I is the main result. Our method wins on three of four targets: ASVspoof2019 (5.03  $\rightarrow$  3.47%, 10/10 seeds), Arabic (22.50  $\rightarrow$  20.99%, 10/10), In-the-Wild (12.78  $\rightarrow$  11.33%, 8/10); on LibriSpeech-TTS it is EER-neutral (33.54  $\rightarrow$  33.74%, improved in only 5/10 seeds), and that is also the target with the lowest source AUC (0.73 vs.  $\geq 0.84$  elsewhere). It is tempting to read source AUC as the precondition, but the apparent  $r=+0.86$  is an artefact of four clusters: within each target the sign is reversed or absent ( $r=-0.77$ ,  $-0.64$ ,  $-0.64$ ,  $+0.12$ ; Fig. 2), and the third-party breadth arm below kills it outright. The precondition that does hold—calibration headroom, and class balance—is established there.

**Significance.** Ten seeds per target make a *per-target* Wilcoxon possible at all ( $n=10$  floor 0.002). Against source-only, ours is significant on ASVspoof2019 and Arabic ( $p=0.0020$ ) and In-the-Wild ( $p=0.0293$ ), not LibriSpeech-TTS ( $p=0.23$ ); Holm correction keeps the first two ( $p_{adj}=0.0078$ ) and makes In-the-Wild nominal (0.0586), with

a bootstrap agreeing. Pooled over all 40 pairs (descriptive only, clustered) ours beats source-only ( $p=2.4\times 10^{-5}$ ), Tent ( $1.5\times 10^{-7}$ ), self-training-only (0.0056) and both DG baselines (30/40 each).

**Tent is not merely unstable, it is unpredictably catastrophic.** On ASVspoof2019 its per-seed EER splits five seeds near source-only and four at or near chance with no warning sign (std 21.96, mean 23.47); on Arabic it is a *consistent* chance-level failure ( $47.35\pm 5.03\%$ , 9/10 seeds). Our method shows no comparable collapse on any target (largest std 2.01). The failure is a ranking failure first: run unmodified on the strongest and weakest public checkpoints below, Tent collapses to chance on one ( $16.41 \rightarrow 49.80\%$  EER, AUC  $.918 \rightarrow .502$ ) [17] but only degrades AUC on the other ( $.986 \rightarrow .979$ ) [16]—an EER collapse appears only where there was AUC headroom to lose.

**Transfer to third-party detectors.** Every result above adapts our own source model, so we apply the published configuration unmodified to nine released detectors we did not train—XLS-R+AASIST checkpoints trained on WaveFake [16] and on ASVspoof2019 [17], wav2vec2 sequence classifiers, and an Audio Spectrogram Transformer [11]—across ten corpora, 369 cells in all.

The consistent effect is *re-calibration*. Writing  $\delta = (100 - \text{EER}) - \text{acc}@0.5$  for the accuracy a model’s ranking supports but its shipped threshold does not deliver, adaptation drives  $\delta$  from 7.29 to 1.96 points (248/369 cells, Wilcoxon  $p=2.2\times 10^{-17}$ ), and  $\text{corr}(\delta_{\text{before}}, \delta_{\text{after}}) = +0.17$ : where it lands barely depends on where it started. Within families it holds wherever there is a deficit to close—the spectrogram transformer 39/39 ( $p=5\times 10^{-8}$ ), the WaveFake-trained XLS-R 39/40, one ASVspoof2019-trained XLS-R ( $p<10^{-11}$ )—is a no-op on families whose shipped threshold is already near-optimal, and *worsens* calibration on one wav2vec2 classifier. That the same LayerNorm-and-head selection rule, designed against XLS-R, re-calibrates a mel-spectrogram transformer unchanged is the clearest architecture-independence evidence here.

Ranking is a different matter. On the 209 cells whose corpora share ASVspoof lineage the EER change is *not* significant ( $+0.28\%$ ,  $p=0.52$ ); adding the two lineage-independent corpora it becomes small and split ( $+0.9\%$ ,  $p=0.003$ )—an order of magnitude under the 5.3-point calibration shift, and per corpus a significant improvement on one, degradation on the other. Our earlier single-corpus result (In-the-Wild, four checkpoints all improved) reproduces but does not generalise—four of 12 ten-seed cells are significant degradations, the largest  $38.13 \rightarrow 41.91\%$ —and it is not memorisation (inductive 4.08 vs. transductive 4.06%).

**A label-free threshold rule captures most of it—but not all.** If the whole benefit were a shifted operating point, gradient descent would be unnecessary. Thresholding each source checkpoint’s pool at the *median* score (label-free) recovers 94% of TTA’s accuracy gain over 70 of our own cells (75.3 vs. 75.6%, oracle 76.6%) and *beats* it over 63 third-party cells (75.6 vs. 73.7%,  $p=0.04$ , where adaptation

degrades a checkpoint’s ranking); pooled over 133 cells the two are indistinguishable, both assuming a class-balanced pool (Sec. V). What a threshold *cannot* do is change ROC-AUC, and where the source ranking is strong the gradient method does: on ASVspoof2019 it raises within-attack-family AUC on all six families (30/30 attack $\times$ seed cells,  $+1.1$  EER,  $p=10^{-9}$ ), largest on the two hardest; on AUC-0.71 LibriSpeech-TTS there is nothing to sharpen and the effect is null. The equivalence holds where the method reduces to re-calibration; given a usable ranking, adaptation adds real separability.

**Where EER gains do occur, they track calibration headroom, and lineage predicts headroom.** Extending our own model past the four original targets: three ASVspoof2021 tracks (LA/DF/PA) erase the pooled EER benefit ( $+0.88 \rightarrow -0.68$ ,  $n=70$ ,  $p=0.80$ )—but these are relatives of ASVspoof2019, which is *in* the source pool, so they arrive already well-thresholded (deficit  $-2.7$  to  $+3.2$ ) with nothing to repair. Two corpora that share no lineage with any source—WaveFake (vocoder artefacts over LJSpeech) [16] and a 2024 commercial-TTS set (ElevenLabs, Polly, Kokoro, Hume, Speechify)—instead arrive badly mis-thresholded (deficit 15.6, 5.8) and adaptation improves their EER by  $+2.5$  and  $+2.9$  points (10 seeds each,  $p=0.002$ ), *larger* than any original target ( $+1.48$  pooled over the six independent targets,  $n=60$ ,  $p<10^{-8}$ ). The nine third-party checkpoints on these two corpora (160 cells) confirm it from the other side: WaveFake ( $\delta=9.0$ ) is repaired and EER improves ( $+6.2\%$ ,  $p=10^{-13}$ ), commercial-TTS ( $\delta=3.4$ ) has nothing to repair and EER drifts worse ( $-2.6\%$ ). The positive EER numbers are contingent not on which targets were to hand but on calibration headroom, which lineage-relatedness predicts.

**The adaptation budget is a confound we introduced.**  $E$  counts epochs, so the number of updates scales with pool size:  $E=4$  over In-the-Wild’s 31,779 clips is  $5.3\times$  the steps of  $E=4$  over a 6,000-clip pool. On our own model at ten seeds,  $E=32$  lifts the Arabic gain from  $+1.41$  to  $+3.65$  EER on 10/10 seeds ( $p=0.002$ ), while on LibriSpeech-TTS—the weakest source ranking—it does not help (2/10). Handed the same  $8\times$  budget the baselines do not benefit (Tent worsens by 5 EER points on Arabic, self-training-only is flat), so the gain is the method’s, not the compute’s. On third-party checkpoints, both significant AST *degradations* reverse at  $E=32$ : EER gain  $-3.78 \rightarrow +7.24$  and  $-2.14 \rightarrow +1.86$ , positive on 6/6 seeds each ( $p=0.031$ ).  $E$  belongs in step counts, not epochs; but more steps is not a free win (it hurts the weakest-ranking target monotonically), so we report the defect rather than re-tabulate around it.

**Backbone and prior DG baselines.** RawNet2Lite (no SSL pretraining) stays near chance on every target (AUC 0.40–0.65) even retrained for 40 epochs, so cross-corpus viability starts from SSL pretraining, not the TTA step. Both train-time DG baselines are beaten pooled over all 40 pairs (30/40 each); per target we beat DANN on two of four and ASDG on three of four (ASDG better on LibriSpeech-TTS). DANN sees the held-out target’s unlabeled audio every step yet is worse, costs

TABLE II

TEST-TIME ADAPTATION VS. TWO TRAIN-TIME DOMAIN-GENERALISATION BASELINES, MEAN EER % / AUC OVER THE SAME 10 SEEDS FOR EVERY METHOD. BEST PER ROW IN **BOLD**. OURS WINS 30/40 (TARGET, SEED) PAIRS AGAINST EACH; PER TARGET WE BEAT ASDG ON THREE OF FOUR AND DANN ON TWO OF FOUR, AND ASDG IS BETTER ON LIBRISPEECH-TTS. ONLY DANN SEES TARGET AUDIO DURING TRAINING. ASDG RAN ON DIFFERENT HARDWARE, SO ITS WALL-CLOCK IS NOT COMPARED.

| Target          | Source       | DANN [8]     | ASDG [5]            | Ours                |
|-----------------|--------------|--------------|---------------------|---------------------|
| ASVspoof2019    | 5.03 / .989  | 7.35 / .966  | 4.98 / .987         | <b>3.47</b> / .994  |
| LibriSpeech-TTS | 33.54 / .727 | 34.43 / .717 | <b>32.64</b> / .749 | 33.74 / .727        |
| In-the-Wild     | 12.78 / .945 | 13.34 / .924 | 17.33 / .891        | <b>11.33</b> / .958 |
| Arabic          | 22.50 / .843 | 28.53 / .759 | 27.41 / .798        | <b>20.99</b> / .866 |

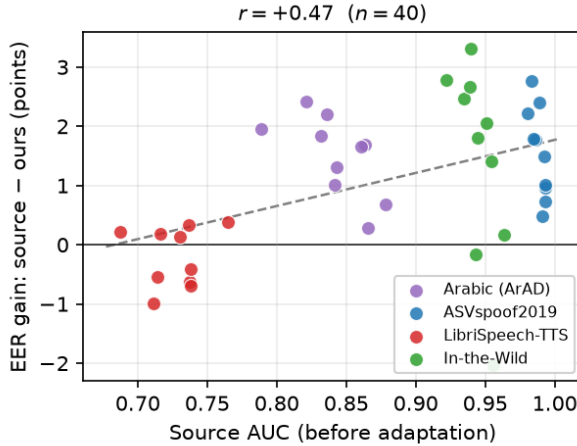


Fig. 2. EER gain (source—ours) vs. source AUC before adaptation, one point per (target, seed), all 40 runs. The association is between corpora, not within them: the four target means give  $r=+0.86$  ( $n=4$ ,  $p=0.14$ ) and the pooled 40-point fit gives  $r=+0.47$ , but within each target the sign is reversed or absent ( $r=-0.77$ ,  $-0.64$ ,  $-0.64$ ,  $+0.12$ ). LibriSpeech-TTS (red, lowest AUC) is the only target with a negative mean gain.

3 $\times$  as much, and is seed-unstable; ASDG beats plain source-only on only two of four (Table II). Neither is reliable here.

**Mechanism.** An inductive check (disjoint adapt/eval halves) rules out transductive memorisation. The proxy  $\mathcal{A}$ -distance and a linear domain-origin probe both *increase* slightly after adaptation (Arabic  $\hat{d}_A$ : 1.545  $\rightarrow$  1.606), the opposite of what domain-invariance predicts: adaptation re-shapes the boundary with rank-reliable pseudo-labels rather than erasing domain identity—consistent with DANN underperforming despite optimising that quantity. MLAAD fake recall is uniform across seen and unseen languages (93.9 vs. 93.3%).

## V. DISCUSSION AND LIMITATIONS

The method is transductive (unlabeled target inputs at adaptation time, the standard TTA setting); an inductive check reproduces every result including the LibriSpeech-TTS null. We report per-target, per-seed numbers throughout, so the limitations below can be read against the evidence:

- **Needs an already-reliable source ranking—and we cannot say how much of that is SSL.** A from-scratch

or low-AUC starting point is not a regime we can rescue (Fig. 2). Our non-SSL backbone never reaches a ranking worth adapting (AUC 0.40–0.65), so it fails the precondition before TTA becomes applicable and we did not run adaptation on it; every third-party checkpoint we adapt is itself a pretrained transformer.

- **Class balance is a second precondition—partly removable.** The fixed-quantile rule labels the bottom  $q$  of the pool “confident real” whatever the true prior, so it targets a *balanced*-prior operating point. A controlled skew sweep confirms the mechanism: at 50  $\rightarrow$  70  $\rightarrow$  90% fake the accuracy gain goes  $+7.6 \rightarrow -3.5 \rightarrow -26.5$  points while ranking stays flat (this also accounts for the ASVspoof2021-DF collapse: source 26.33% EER, TTA 42.46% on a 97%-spoof pool). But the prior is estimable label-free by black-box shift estimation ( $\hat{p}=M^{-1}q$ ,  $M$  the source confusion matrix): across 50–95% fake, BBSE recovers it to within 1.4% and re-pointing the threshold at the  $\hat{p}$  quantile matches the *labelled* accuracy-optimal threshold to within 0.4 points—a score GMM does not (28% error at 90% fake)—and it is robust to moderate calibration drift, reading source error structure not the target score shape. So the threshold is recoverable at any prior; the confident-tail pseudo-labels stay noisy under severe skew even with a prior-matched budget. Our source model is also not state-of-the-art (26.33% vs. published 8–9% on 2021-DF), so this is a relative-gain claim on a fixed backbone.
- **Fixed hyperparameters.**  $q$  and  $\lambda$  are label-free and not per-target optimal; a label-free  $\lambda$  schedule pays nothing at our scale (within 0.24 EER of a per-point oracle).  $E$  is not a hyperparameter but a mis-specified budget, as above.
- **Scale.** The main table is four targets, ten seeds; going from three to ten seeds made the per-target tests possible but shrank our DG margin to two per-target wins against DANN and three against ASDG. Breadth comes from the third-party arm (nine detectors, ten corpora, 369 cells), where the source-AUC ordering was falsified and the calibration account confirmed. Two large HF checkpoints run at batch 8 not 16, and three of the nine have undocumented training data and sit in a separate arm.

## VI. CONCLUSION

Cross-corpus audio deepfake detection fails because the operating point is mis-calibrated, not because ranking is lost. Our unsupervised TTA repairs it (369 third-party cells,  $p \approx 10^{-17}$ , ranking effect an order of magnitude smaller), and a label-free median-threshold move matches it on average. The gradient step earns its cost only where the source ranking is strong—on ASVspoof2019 it sharpens within-attack ROC-AUC (30/30 cells), which no threshold can—and the class-balance requirement is largely removable by label-free prior estimation. Source AUC does not predict the gain and is retired. Given a usable source ranking the method re-calibrates and modestly re-separates; without one it is a threshold rule.

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